Digital Integrated Circuit Design

Lecture-1: Review of CMOS Digital Circuits

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Outline

- ❑ Response of CMOS Inverter
- ❑ Propagation Delay
- ❑ Critical Path and Maximum Operating Frequency

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- ❑ **Propagation Delay**
- ❑ Critical Path and Maximum Operating Frequency

Analysis of CMOS Inverter

- ❑ Two types of analyses for CMOS inverter:
  - ■ **DC analysis** to determine V_{out} for a given value of V_{in}
    - • Voltage transfer characteristic (VTC)
  - ■ **Transient analysis** to show input voltage as function of time corresponding to output voltage
    - • **Delay** between a change in I/P and a corresponding change in O/P
  - ■ PMOS : $\beta_p = k'_p \left(\frac{W}{L}\right)_p, V_{T_p} < 0;\ k'_p = \mu_p C_{ox}$
  - ■ NMOS : $\beta_n = k'_n \left(\frac{W}{L}\right)_n, V_{T_n} > 0;\ k'_n = \mu_n C_{ox}$

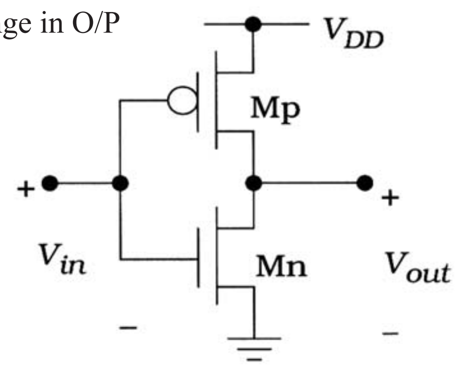
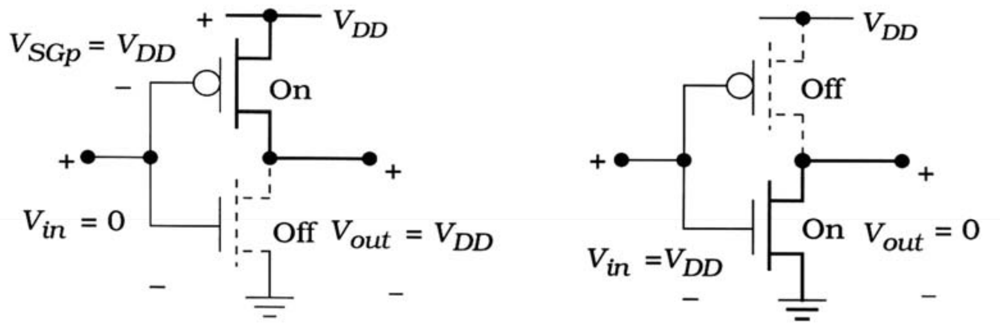


Figure 7.1 The CMOS inverter circuit

DC Response

$V_{OH}=V_{DD}$ $V_{OL}=0$



(a) Low input voltage (b) High input voltage

Figure 7.2 V_{OH} and V_{OL} for the inverter circuit

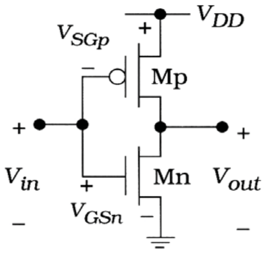
- Logic swing : $V_L=V_{OH}-V_{OL}=V_{DD}$ (Full rail output)

Voltage Transfer Characteristic (VTC)

- Plot of V_{out} as a function of V_{in}
- V_{in} from ground (0V) up to power supply voltage (V_{DD})
  - NMOS cut off: Vin < V_{Tn}
  - PMOS cut off: Vin > VDD-$|V_{Tp}|$
  - $V_{IL}=V_{Tn}=$ and $V_{IH}=V_{DD}-|V_{Tp}|$
- Logic swing V_L
  - $V_L=V_{OH}-V_{OL}=V_{DD}$

$V_{SGp}=V_{DD}-V_{in}$

$V_{GSn}=V_{in}$



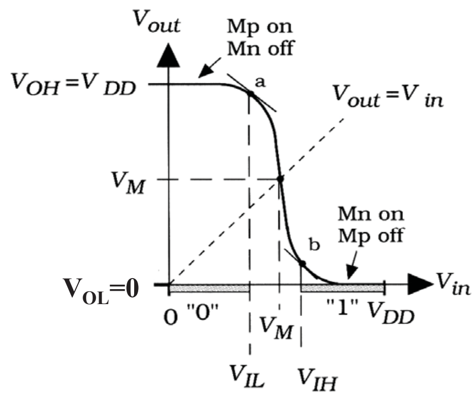


Figure 7.3 Voltage transfer curve for the NOT gate

Voltage Transfer Characteristic (cont'd)

- "a" point:
  - Slop = -1 and define as input low voltage V_{IL}
  - $0 \le V_{in} \le V_{IL}$ and define as logic 0
- "b" point:
  - Slop = -1 and define as input high voltage V_{IH}
  - $V_{IH} \le V_{in} \le VDD$ and define as logic 1
- Voltage noise margin

$$V_{NM_H} = V_{OH} - V_{IH}$$

$$V_{NM_L} = V_{IL} - V_{OL}$$

  - Measure how stable of the inputs

$\boxed{V_M = V_{in} = V_{out}}$

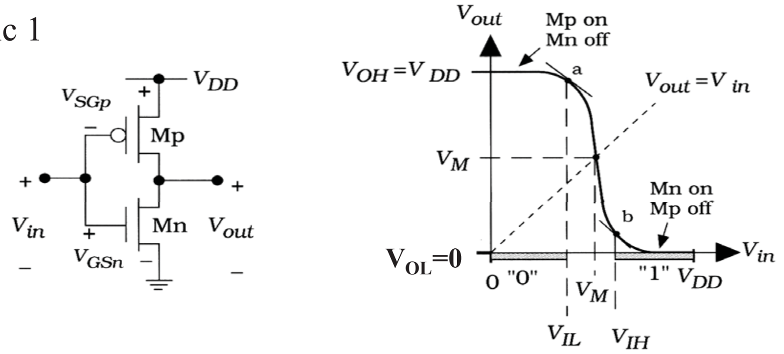
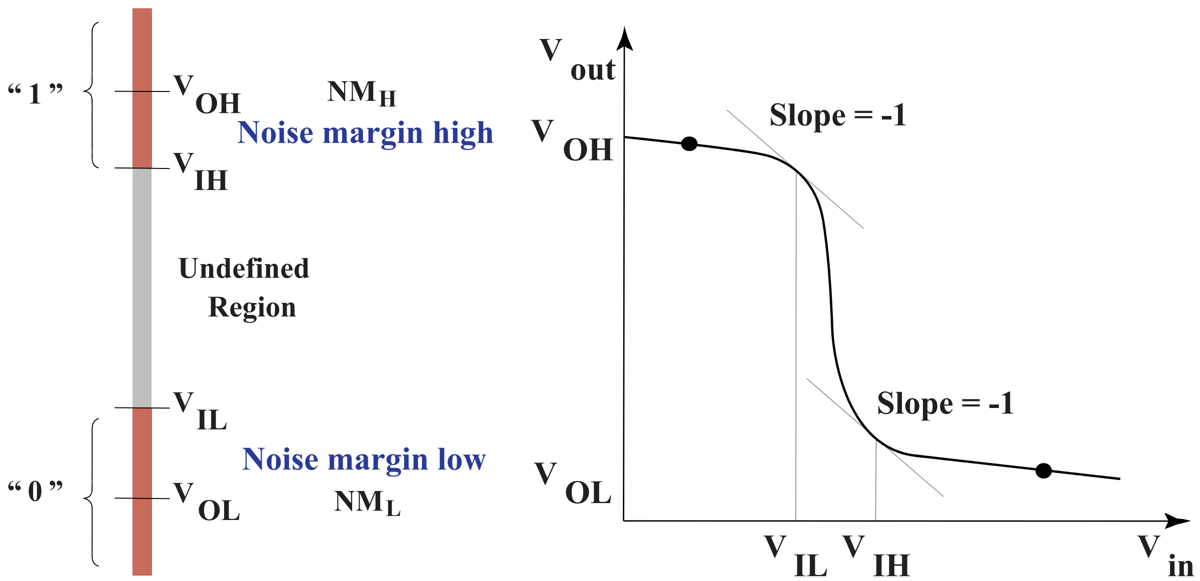


Figure 7.3 Voltage transfer curve for the NOT gate

Mapping between Analog and Digital Signals



V_M Value

- Define logic 0 and 1 input voltage to calculate the exact voltage V_M
- V_M=Vin=Vout and $I_{Dp}=I_{dn}$
  - For nMOS, $V_{sat}=V_{GSn}-V_{Tn}=V_M-V_{Tn}$;
  - For pMOS, $V_{sat}=V_{SGp}-|V_{Tp}|=V_{DD}-V_M-|V_{Tp}|$
- Need to know which operating region (sat or non-sat) to calculate current
  - V_{DSn}=Vout=V_M → V_{DSn} > Vsat @ sat region
  - $V_{DSp}=V_{DD}-V_M$ → V_{DSp} > Vsat @ sat region
  - $I_{D_n} = \frac{\beta_n}{2}\left(V_{GS_n} - V_{T_n}\right)^2$
  - $I_{D_p} = \frac{\beta_p}{2}\left(V_{SG_p} - \left|V_{T_p}\right|\right)^2$

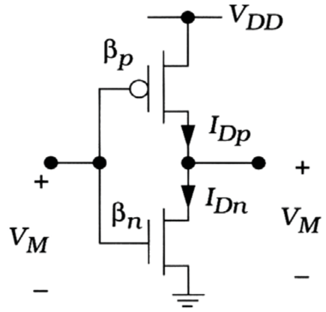


Figure 7.4 Inverter voltages for V_M calculation

V_M Value (cont'd)

- $I_{D_p} = I_{D_n}$

$$\frac{\beta_p}{2}\left(V_{SG_p} - \left|V_{T_p}\right|\right)^2 = \frac{\beta_n}{2}\left(V_{GS_n} - V_{T_n}\right)^2$$

$$\frac{\beta_p}{2}\left(V_{DD} - V_{\mathrm{M}} - \left|V_{T_p}\right|\right)^2 = \frac{\beta_n}{2}\left(V_{\mathrm{M}} - V_{T_n}\right)^2$$

$$\left(V_{DD} - V_{\mathrm{M}} - \left|V_{T_p}\right|\right) = \sqrt{\frac{\beta_n}{\beta_p}}\left(V_{\mathrm{M}} - V_{T_n}\right)$$

$$V_{\mathrm{M}} = \frac{V_{DD} - \left|V_{T_p}\right| + \sqrt{\frac{\beta_n}{\beta_p}}V_{T_n}}{1 + \sqrt{\frac{\beta_n}{\beta_p}}}$$

V_M Value (cont'd)

- Relationship of β_p and β_n

$$\frac{\beta_n}{\beta_p} = \frac{k'_n\left(\frac{W}{L}\right)_n}{k'_p\left(\frac{W}{L}\right)_p} = 1 \rightarrow \frac{k'_n}{k'_p} = \frac{\mu_n}{\mu_p} = \gamma = \frac{\left(\frac{W}{L}\right)_p}{\left(\frac{W}{L}\right)_n}$$

  - $k'_p = \mu_p C_{ox}$ and $k'_n = \mu_n C_{ox}$

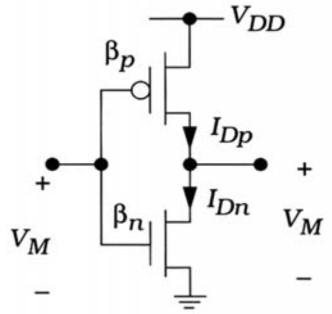


Figure 7.4 Inverter voltages for V_M calculation

V_M Value (cont'd)

- Symmetrical inverter VTC
  - Equal "0" and "1" input ranges
  - $V_M = \frac{V_{DD}}{2} \rightarrow \frac{\beta_n}{\beta_p} = \left(\frac{\frac{V_{DD}}{2} - |V_{Tp}|}{\frac{V_{DD}}{2} - V_{Tn}}\right)^2$
  - Allow us to compute the transistor size for this particular choice of V_M
  - Let $V_{Tn} = |V_{Tp}|$, then $\beta_n = \beta_p$
- MOS size
  - $\frac{k'_n}{k'_p} = \frac{\mu_n}{\mu_p} = \gamma \rightarrow \frac{\left(\frac{W}{L}\right)_n}{\left(\frac{W}{L}\right)_p} = \frac{k'_p}{k'_n} = \frac{1}{\gamma}$
- Device Ratio Dependency on V_M

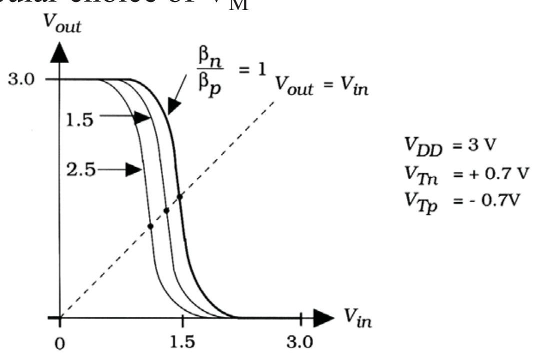


Figure 7.6 Dependence of V_M on the device ratio

Example

Example 7.1
Consider a CMOS process with the following parameters

$$k'_n = 140 \ \mu A/V^2 \qquad V_{Tn} = +0.70 \ \text{V}$$
$$k'_p = 60 \ \mu A/V^2 \qquad V_{Tp} = -0.70 \ \text{V} \tag{7.21}$$

with $V_{DD} = 3.0$ V.

Consider the case where $\beta_n = \beta_p$. We can verify that this is a symmetrical design by calculating

$$V_M = \frac{3 - 0.7 + \sqrt{1}(0.7)}{1 + \sqrt{1}} = 1.5 \text{ V} \tag{7.22}$$

so that V_M is one-half the value of the power supply voltage. To achieve this design, we must choose the device aspect ratios such that

$$\frac{\beta_n}{\beta_p} = \frac{k'_n\left(\frac{W}{L}\right)_n}{k'_p\left(\frac{W}{L}\right)_p} = 1 \tag{7.23}$$

where we recall that the process transconductance parameters k' are given by $k' = \mu_n C_{ox}$, and are set by the processing. For the present case, we rearrange the expression to read

$$\left(\frac{W}{L}\right)_p = \frac{k'_n}{k'_p}\left(\frac{W}{L}\right)_n \tag{7.24}$$

$$\left(\frac{W}{L}\right)_p = \left(\frac{140}{60}\right)\left(\frac{W}{L}\right)_n = 2.33\left(\frac{W}{L}\right)_n \tag{7.25}$$

This shows that the pFET must be about 2.33 times larger than the nFET.

Let us now examine the case where the nFET and the pFET have the same aspect ratio: $(W/L)_n = (W/L)_p$. With the values provided in the problem statement,

$$\frac{\beta_n}{\beta_p} = \frac{k'_n}{k'_p} = 2.33 \tag{7.26}$$

so that the midpoint voltage is given by

$$V_M = \frac{3 - 0.7 + \sqrt{2.33}\ (0.7)}{1 + \sqrt{2.33}} = 1.33 \text{ V} \tag{7.27}$$

This choice shifts V_M to a value that is smaller than $(V_{DD}/2)$.

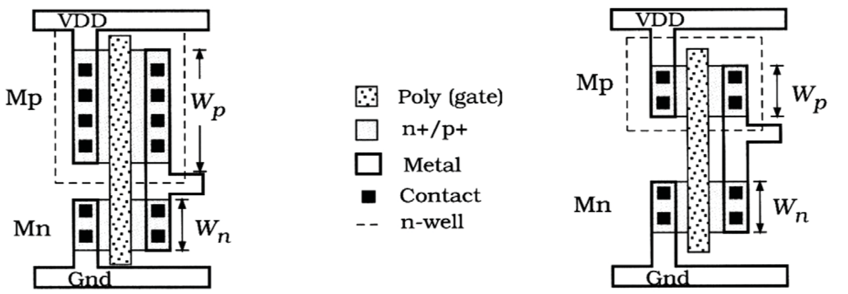
$$V_M = \frac{V_{DD} - |V_{T_p}| + \sqrt{\frac{\beta_n}{\beta_p}} V_{T_n}}{1 + \sqrt{\frac{\beta_n}{\beta_p}}} \qquad \frac{\beta_n}{\beta_p} = \frac{k'_n\left(\frac{W}{L}\right)_n}{k'_p\left(\frac{W}{L}\right)_p}$$

Example (cont'd)

- Layout of CMOS Inverter
  - What is effect of aspect ratio on the VTC?

$W_p = 2W_n$, $V_M = \frac{1}{2} V_{DD}$

$W_p = W_n$, $V_M < \frac{1}{2} V_{DD}$



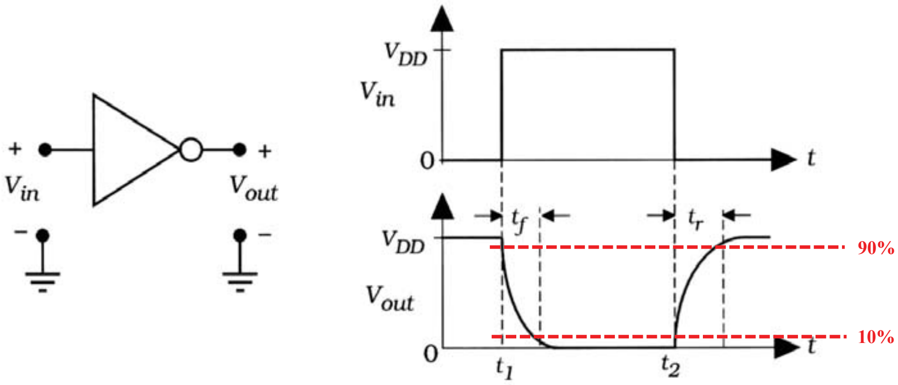
(a) Larger pFET design (b) Equal aspect ratios

Figure 7.5 Comparison of the layouts for Example 7.1

- Increase (β_n/β_p) → decrease V_M

Inverter Switching Characteristics

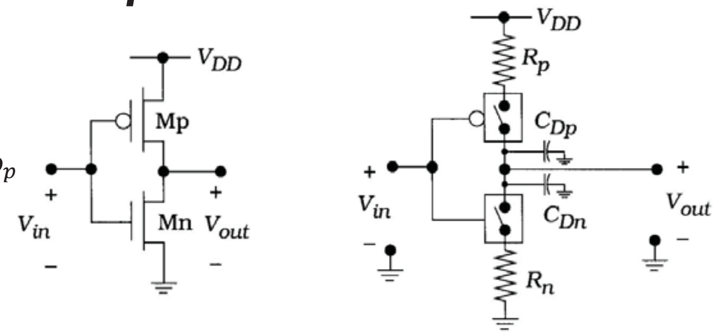
- Minimum amount of time delay when the inputs change
- ***Fall time*** t_f when output change from 1 to 0
- ***Rise time*** t_r when output change from 0 to 1



- t_f and t_r delay due to parasitic ***R*** and parasitic ***C*** of transistor

RC Model of CMOS Inverter

- Operating in tridode region
  - $I_{D_n} = \beta_n (V_{GS_n} - V_{T_n})\, V_{DS_n}$
  - $I_{D_p} = \beta_p \left(V_{SG_p} - |V_{T_p}|\right) V_{SD_p}$



(a) FET circuit (b) RC switch model equivalent

- Parasitic resistance and parasitic capacitance

$$R_n = \frac{1}{\beta_n (V_{DD} - V_{T_n})} \qquad C_{Dn} = C_{GSn} + C_{DBn} = \frac{1}{2} C_{ox} L' W_n + C_{jn} A_n + C_{jswn} P_n$$

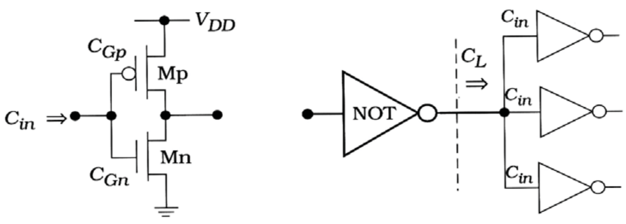
$$R_p = \frac{1}{\beta_p (V_{DD} - |V_{Tp}|)} \qquad C_{Dp} = C_{GSp} + C_{DBp} = \frac{1}{2} C_{ox} L' W_p + C_{jp} A_p + C_{jswp} P_p$$

RC Model of CMOS Inverter with Load

- In logic chain, next logic gate definitely driven by previous logic gate
- Define number of gates driven by the previous stage as **fan-out**
- Define C_{in} as input cap. of logic gate
- Define C_L as output cap. of logic gate and equal to input cap. of next stage

$$\boxed{C_{in} = C_{Gp} + C_{Gn}}$$

$$\boxed{\begin{aligned} C_L &= 3C_{in} \\ &= 3(C_{Gp} + C_{Gn}) \end{aligned}}$$



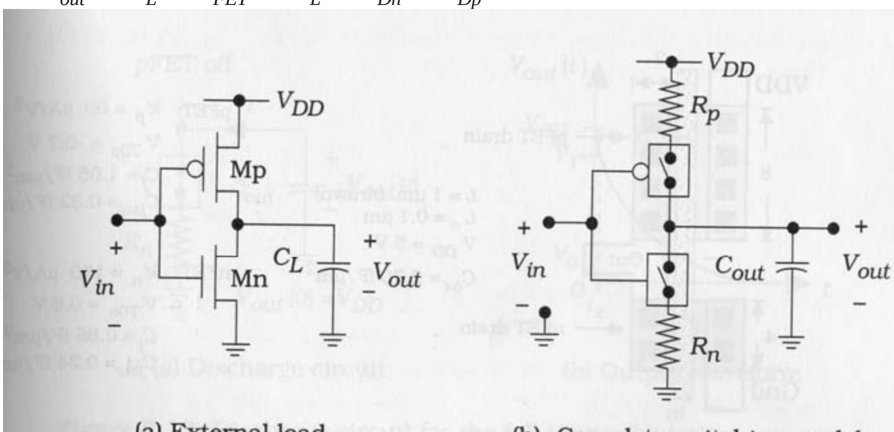
(a) Single stage (b) Loading due to fan-out

Input capacitance and load effects

RC Model of Inverter with Load

- Total load for the output node

$$C_{out} = C_L + C_{FET} = C_L + C_{Dn} + C_{Dp}$$



(a) External load (b) Complete switching model

Evolution of the inverter switching model

Fall Time Calculation

- Discharge C_{out} from VDD to 0

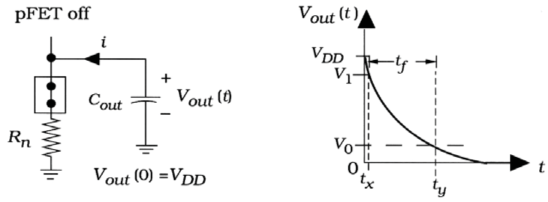
$$i = -C_{out}\frac{dV_{out}}{dt} = \frac{V_{out}}{R_n}$$

- Initial condition $V_{out}(0)=V_{DD}$

$$V_{out}(t) = V_{DD}e^{-t/\tau_n} \qquad \tau_n = R_n C_{out} \quad (time\ \ constant)$$

- Fall time: **$0.9V_{DD}$ to $0.1V_{DD}$**

$$t = \tau_n \ln\left(\frac{V_{DD}}{V_{out}}\right)$$



(a) Discharge circuit (b) Output waveform

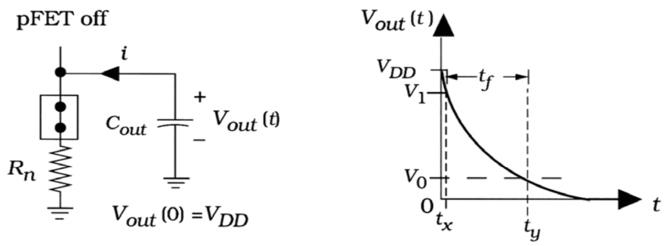
Discharge circuit for the fall time calculation

Fall Time Calculation (cont'd)

$$t_f = t_y - t_x = \tau_n\, ln\left(\frac{V_{DD}}{0.1\cdot V_{DD}}\right) - \tau_n\, ln\left(\frac{V_{DD}}{0.9\cdot V_{DD}}\right) = \tau_n\, ln(9)$$

$$t_{HL} = t_f \approx 2.2\tau_n$$

$$\ln(a) - \ln(b) = \ln\left(\frac{a}{b}\right)$$



(a) Discharge circuit (b) Output waveform

Discharge circuit for the fall time calculation

Rise Time Calculation

- Charge C_{out} from 0 to V_{DD}

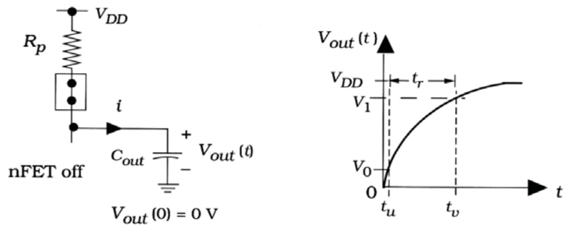
$$i = C_{out}\frac{dV_{out}}{dt} = \frac{V_{DD} - V_{out}}{R_p}$$

- Initial condition $V_{out}(0)=0$

$$V_{out}(t) = V_{DD}\left(1 - e^{-t/\tau_p}\right) \qquad \tau_p = R_p C_{out} \quad (time\ \ constant)$$

- Rise time: $0.1V_{DD}$ to $0.9V_{DD}$

$$t_{LH} = t_v - t_u$$
$$= t_r \approx 2.2\tau_p$$



(a) Charge circuit (b) Output waveform

Rise time calculation

Maximum Operating Frequency

- Meaning of t_{LH} and t_{HL}
  - Shortest time need for output change from logic-0 to logic-1 or from logic-1 to logic-0
- 50% duty cycle of clock
  - Half period with voltage 0 and half period with VDD
- Meaning of max. signal frequency
  - Largest frequency applied to gate and allow output to settle in a definable state
- Signal frequency > f_{max}
  - Not have sufficient time to stabilize the correct value of output voltage for logic gate

$$f_{\max} = \frac{1}{t_{HL} + t_{LH}}$$

- Signal frequency ≤ f_{max}
  - Allow the output of the logic gate to settle in a correct state

Propagation Delay

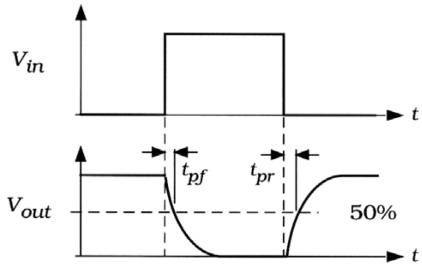
- Propagation delay time t_p
  - Estimate the reaction delay time from input to output
  - t_{pf} : fall time from maximum level to $0.5V_{DD}$ (50%)
  - t_{pr} : rise time from 0 to $0.5V_{DD}$

$$t_p = \frac{t_{pf} + t_{pr}}{2} \qquad t_{pf} = \ln(2)\tau_n \qquad \tau_n = R_n C_{out}$$

$$t_{pr} = \ln(2)\tau_p \qquad \tau_p = R_p C_{out}$$

$$t_p \approx 0.35(\tau_n + \tau_p)$$

$$\boxed{t_p = 0.693\, C_L\, (R_n + R_p)/2}$$

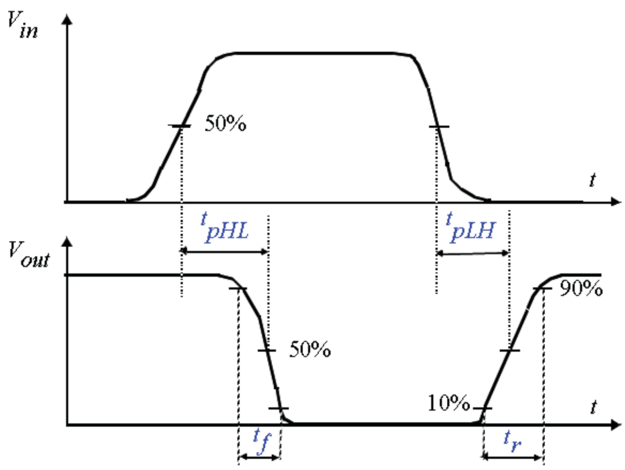


Propagation time definitions

- Propagation delay
  - Commonly used in basic logic simulation programs

Propagation Delay of a Logic Gate

- Propagation delay of a logic gate
  - The ***delay experienced*** by a signal when passing through ***a gate***
  - Measure ***between 50% transition points*** of the input and output waveform
  - $t_p = \frac{t_{pHL} + t_{pLH}}{2}$



- Fall and rise time
  - t_{pf}
  - t_{pr}

Propagation Delay of a Logic Gate (cont'd)

- Propagation delay t_{pd}
  - The **maximum** time from the input crossing 50% to the output crossing 50%
- Contamination delay t_{cd}
  - The **minimum** time from the input crossing 50% to the output crossing 50%

Outline

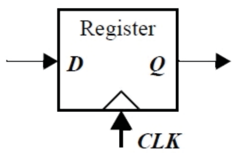
- Propagation Delay
- **Critical Path and Maximum Operating Frequency**

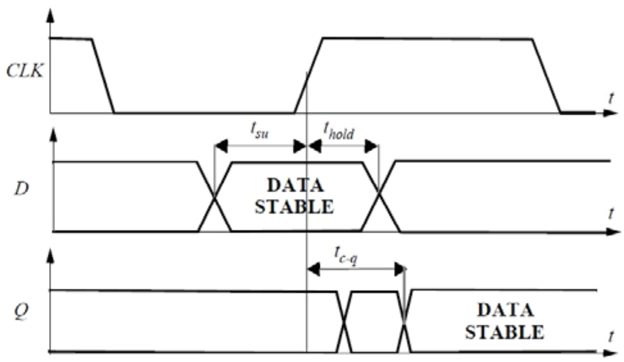
Critical Path

- Definition
  - The **longest path** through the logic, which determines the ultimate speed of the structure, is called the critical path.
  - The **maximum propagation delay** of a logic gate (or a combinational logic)

Set-Up and Hold Time of a Register

- Register (D-type FF)
  - Set-up time (t_{su}): The data input must be valid before the CLK transition.
  - Hold time (t_{hold}): The input must remain valid after the CLK edge.
  - Clock-to-Q delay ($t_{c\text{-}q}$)





Set-up and Hold Time of a Register (cont'd)

- The minimum clock period T_{CLK} (The maximum operating frequency f_{CLK}) for a pipelined system
  - $T_{CLK} = t_{c-q} + t_{plogic} + t_{su}$
  - $f_{CLK} = \frac{1}{T_{CLK}}$